PART A: EMPLOYABILITY SKILLS

(Refer to IT 402 notes for detailed coverage of Employability Skills)

UNIT 1: COMMUNICATION SKILLS-II
- Verbal, Non-Verbal, Written, Visual Communication
- Communication Styles: Passive, Aggressive, Assertive, Passive-Aggressive
- Barriers to Communication: Physical, Language, Psychological, Perceptual, Cultural, Organisational
- Active Listening, Empathy, Feedback
- Email Writing and Formal Writing

UNIT 2: SELF-MANAGEMENT SKILLS-II
- Self-Awareness, Self-Regulation, Self-Motivation
- Time Management: Prioritisation, To-Do Lists, Deadlines
- SMART Goals: Specific, Measurable, Achievable, Relevant, Time-bound
- Stress Management: Exercise, Breathing, Time Management, Hobbies

UNIT 3: ICT SKILLS-II
- ICT Components: Hardware, Software, Networks
- Digital Devices: Computers, Laptops, Tablets, Smartphones
- Internet: Browsers, Search Engines, Email
- Digital Safety: Antivirus, Firewall, Strong Passwords
- Cyber Threats: Virus, Malware, Phishing, Hacking

UNIT 4: ENTREPRENEURIAL SKILLS-II
- Entrepreneurship: Starting and running a business
- Types of Business: Micro, Small, Large
- Business Plan: Executive Summary, Market Analysis, Financial Plan
- Financial Literacy: Revenue, Expenses, Profit, Loss, Cash Flow
- Risk Management: Identify, Reduce, Transfer, Accept

UNIT 5: GREEN SKILLS-II
- Sustainable Development: Meeting present needs without compromising future
- Green Economy: Low-carbon, resource-efficient, socially inclusive
- Green Jobs: Solar panel installer, wind turbine technician, environmental engineer
- Natural Resources: Renewable (solar, wind) and Non-Renewable (coal, oil)
- Environmental Issues: Climate Change, Pollution, Deforestation
- Sustainable Practices: Energy Conservation, Water Conservation, Waste Management (3Rs)

PART B: SUBJECT SPECIFIC SKILLS

UNIT 1: REVISITING AI PROJECT CYCLE & ETHICAL FRAMEWORKS FOR AI

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1. AI Project Cycle

Definition: A systematic framework for developing AI projects from problem identification to deployment.

Stages of AI Project Cycle:

A. Problem Scoping:
- Identifying the problem to be solved.
- Setting goals and objectives.
- Understanding the context and stakeholders.
- Determining success criteria.

B. Data Acquisition:
- Collecting relevant data.
- Identifying data sources (primary/secondary).
- Ensuring data quality and reliability.
- Addressing data privacy and ethical concerns.

C. Data Exploration:
- Understanding the data.
- Visualising data patterns.
- Identifying outliers and missing values.
- Analysing data distribution.

D. Modeling:
- Selecting appropriate AI models.
- Training models on data.
- Testing model performance.
- Iterative refinement.

E. Evaluation:
- Assessing model accuracy.
- Validating results.
- Identifying biases and errors.
- Making improvements.

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2. The Three Domains of AI

A. Data:
- The foundation of AI.
- Collecting, processing, and analysing data.
- Data visualisation.
- Data-driven decision making.

B. Computer Vision:
- Enabling machines to see and interpret visual information.
- Applications: Image recognition, object detection, face recognition.

C. Natural Language Processing (NLP):
- Enabling machines to understand and process human language.
- Applications: Chatbots, language translation, sentiment analysis.

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3. Ethical Frameworks for AI

Definition: Guidelines and principles that govern the ethical development and use of AI systems.

Need for Ethical Frameworks:
- Prevent AI bias and discrimination.
- Ensure transparency and accountability.
- Protect privacy and data security.
- Promote fairness and social good.

Types of Ethical Frameworks:

A. Deontological Ethics:
- Focus on duties and rules.
- Example: Following a code of conduct.

B. Consequentialist Ethics:
- Focus on outcomes and consequences.
- Example: Maximising good outcomes.

C. Virtue Ethics:
- Focus on character and moral virtues.
- Example: Acting with honesty and integrity.

D. Bioethics:
- Ethical framework used in healthcare.
- Focus on patient rights, informed consent, and confidentiality.
- **Case Study:** Using AI in medical diagnosis – ensuring accuracy and avoiding misdiagnosis.

Ethical Issues in AI:
- **Bias:** AI models can reflect societal biases.
- **Transparency:** Lack of understanding of how AI makes decisions.
- **Accountability:** Who is responsible for AI decisions?
- **Privacy:** Protecting personal data.
- **Job Displacement:** Automation replacing human jobs.

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. What are the stages of the AI Project Cycle?
2. What are the three domains of AI? Give examples.
3. Why are ethical frameworks important in AI?
4. What is bioethics? Give an example.

Long Answer (4-5 marks):
1. Explain the AI Project Cycle with examples.
2. Describe the ethical issues related to AI and how they can be addressed.
3. Discuss the importance of ethical frameworks in the development of AI systems.

UNIT 2: ADVANCED CONCEPTS OF MODELING IN AI

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1. Revisiting AI, ML, and DL

A. Artificial Intelligence (AI):
- The broad field of creating machines that can perform tasks that typically require human intelligence.
- **Examples:** Problem-solving, reasoning, perception, language understanding.

B. Machine Learning (ML):
- A subset of AI where machines learn from data without being explicitly programmed.
- **Types:**

  - **Supervised Learning:** Learning from labelled data (input-output pairs).
  - **Unsupervised Learning:** Learning from unlabelled data (finding patterns).
  - **Reinforcement Learning:** Learning through trial and error (rewards and punishments).

C. Deep Learning (DL):
- A subset of ML using neural networks with many layers.
- **Examples:** Image recognition, speech recognition, natural language processing.

Common Terminologies:
- **Feature:** An individual measurable property of the data.
- **Label:** The output or target variable in supervised learning.
- **Training Data:** Data used to train the model.
- **Testing Data:** Data used to evaluate the model.
- **Model:** The mathematical representation learned from data.

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2. Modeling

Types of AI Models:

A. Rule-Based Approach:
- Models based on predefined rules.
- **Example:** A spam filter using rules (if email contains "click here" → spam).

B. Learning-Based Approach:
- Models that learn from data.
- **Example:** A spam filter that learns from labelled emails.

Categories of Machine Learning Models:

A. Supervised Learning:
- **Definition:** Learning from labelled data.
- **Examples:** Classification, Regression.
- **Subcategories:**
  - **Classification:** Predicting a category (e.g., spam or not spam).
  - **Regression:** Predicting a continuous value (e.g., house price).

B. Unsupervised Learning:
- **Definition:** Learning from unlabelled data.
- **Examples:** Clustering, Association.
- **Subcategories:**
  - **Clustering:** Grouping similar data points (e.g., customer segmentation).
  - **Association:** Finding relationships (e.g., market basket analysis).

C. Reinforcement Learning:
- **Definition:** Learning through trial and error.
- **Example:** Game playing (AI learns to play chess).

Subcategories of Deep Learning:
- **Artificial Neural Networks (ANN):** Basic neural network architecture.
- **Convolutional Neural Networks (CNN):** Used for image processing.

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3. Artificial Neural Networks (ANN)

Definition: A computing system inspired by the human brain, composed of interconnected nodes (neurons) that process information.

Structure of ANN:
- **Input Layer:** Receives the input data.
- **Hidden Layers:** Process the data (multiple layers possible).
- **Output Layer:** Produces the final output.

How Neural Networks Learn:
1. **Forward Propagation:** Data flows from input to output.
2. **Loss Calculation:** Error is calculated.
3. **Backpropagation:** Error is propagated backward to adjust weights.
4. **Weight Update:** Weights are adjusted to reduce error.
5. **Iterative Process:** Repeated until the model is accurate.

Key Concepts:
- **Neuron:** A basic unit of computation.
- **Weight:** A number that adjusts the strength of a connection.
- **Bias:** A number added to the input.
- **Activation Function:** Determines if a neuron fires (e.g., sigmoid, ReLU).

Applications:
- Image recognition.
- Speech recognition.
- Natural language processing.
- Autonomous vehicles.

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. Differentiate between AI, ML, and DL.
2. What is the difference between supervised and unsupervised learning?
3. What is a neural network?
4. What is the role of weights in a neural network?

Long Answer (4-5 marks):
1. Explain the different types of machine learning models with examples.
2. Describe the architecture of an artificial neural network.
3. Explain the process of how a neural network learns.

UNIT 3: EVALUATING MODELS

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1. Importance of Model Evaluation

Why Evaluate Models?
- To measure performance.
- To identify weaknesses.
- To compare different models.
- To ensure reliability and accuracy.

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2. Train-Test Split

Definition: Dividing the dataset into training and testing subsets.

Why Split?
- **Training Set:** Used to train the model.
- **Testing Set:** Used to evaluate the model on unseen data.

Common Split: 80% training, 20% testing.

Purpose: To check if the model generalises well to new data.

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3. Accuracy and Error

A. Accuracy:

- **Definition:** The proportion of correct predictions.
- **Formula:** (Correct Predictions) / (Total Predictions) × 100
- **Example:** If the model makes 90 correct predictions out of 100, accuracy is 90%.

B. Error:
- **Definition:** The proportion of incorrect predictions.
- **Formula:** (Incorrect Predictions) / (Total Predictions) × 100
- **Example:** If the model makes 10 incorrect predictions out of 100, error is 10%.

Types of Errors:
- **False Positive (Type I):** Predicting positive when it is actually negative.
- **False Negative (Type II):** Predicting negative when it is actually positive.

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4. Evaluation Metrics for Classification

A. Confusion Matrix:
A table that summarises the performance of a classification model.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

B. Precision:
- **Definition:** Proportion of positive predictions that were actually correct.
- **Formula:** TP / (TP + FP)
- **High Precision:** Few false positives.

C. Recall (Sensitivity):
- **Definition:** Proportion of actual positives that were correctly identified.
- **Formula:** TP / (TP + FN)
- **High Recall:** Few false negatives.

D. F1 Score:
- **Definition:** Harmonic mean of precision and recall.
- **Formula:** 2 × (Precision × Recall) / (Precision + Recall)
- **Balances precision and recall.**

E. Accuracy:
- **Definition:** Overall proportion of correct predictions.
- **Formula:** (TP + TN) / (TP + TN + FP + FN)

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5. Ethical Concerns Around Model Evaluation

A. Bias:
- Models can be biased if training data is biased.
- **Example:** A hiring model trained on biased data may discriminate.

B. Transparency:
- Understanding how models make decisions.
- **Black Box Problem:** Some models are too complex to interpret.

C. Accuracy:
- Overemphasis on accuracy can lead to overlooking errors.
- **Example:** In medical diagnosis, false negatives are more serious than false positives.

D. Fairness:
- Ensuring models treat all groups equally.
- **Example:** A loan approval model should not discriminate based on race or gender.

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. What is the train-test split? Why is it used?
2. Define accuracy and error.
3. What is a confusion matrix?
4. What is the F1 score?

Long Answer (4-5 marks):
1. Explain the different evaluation metrics for classification models.
2. Describe the ethical concerns around model evaluation.
3. What is the difference between precision and recall? When would you use each?

UNIT 4: STATISTICAL DATA (PRACTICAL)

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1. Introduction to Statistical Data

Definition: Data that is collected, analysed, and interpreted using statistical methods.

Types of Data:
- **Quantitative Data:** Numerical data (e.g., height, weight, income).
- **Qualitative Data:** Categorical data (e.g., colour, gender, city).

Importance of Statistics in AI:
- Understanding data distribution.
- Identifying patterns and trends.
- Making predictions.
- Evaluating model performance.

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2. No-Code AI Tools

Definition: Tools that allow users to build AI models without writing code.

Advantages:
- Accessible to non-programmers.
- Rapid prototyping.
- Visual interface.

No-Code vs Low-Code:
- **No-Code:** No programming required (e.g., Orange, Lobe, Teachable Machine).
- **Low-Code:** Minimal coding required (e.g., AutoML tools).

Popular No-Code Tools:
- **Orange Data Mining:** Visual programming for data analysis.
- **Lobe:** Build image classification models.
- **Teachable Machine:** Train models for images, sounds, and poses.

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3. Orange Data Mining Tool

What is Orange?
- An open-source data visualisation and analysis tool.
- Uses a visual programming interface (drag and drop).

Features:
- Data exploration and visualisation.
- Machine learning algorithms.
- Interactive widgets.
- No coding required.

Key Widgets:
- **File:** Load data from CSV, Excel, etc.
- **Data Table:** View data.
- **Scatter Plot:** Visualise data distribution.
- **Distributions:** Analyse data distribution.
- **Predictions:** Evaluate model predictions.

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4. Statistical Data Use Case Walkthrough

Case Study: Palmer Penguins Prediction

Problem: Predict the species of a penguin based on its physical characteristics.

Steps:
1. **Data Loading:** Load the Palmer Penguins dataset.
2. **Data Exploration:** Use scatter plots and distributions to understand the data.
3. **Data Preprocessing:** Handle missing values, encode categorical variables.
4. **Model Selection:** Choose a classification model (e.g., Decision Tree, Random Forest).
5. **Training:** Train the model on the training data.
6. **Evaluation:** Evaluate the model using accuracy, confusion matrix, and other metrics.
7. **Prediction:** Use the model to predict the species of new penguins.

Key Tools Used:
- Orange Data Mining for visual workflow.
- MS Excel for basic statistical analysis.

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. What is statistical data?
2. What is the difference between no-code and low-code AI?
3. What are some popular no-code AI tools?
4. What is Orange Data Mining?

Long Answer (4-5 marks):
1. Explain the steps involved in a data science project using Orange Data Mining.
2. How can statistics be used in AI?
3. Describe the Palmer Penguins case study and how it can be solved using no-code AI.

UNIT 5: COMPUTER VISION (THEORY + PRACTICAL)

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1. Introduction to Computer Vision

Definition: A field of AI that enables computers to derive meaningful information from visual inputs (images, videos).

Goal: To replicate the human visual system.

Applications:
- Image recognition.
- Object detection.
- Face recognition.
- Autonomous vehicles.
- Medical image analysis.
- Augmented reality.

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2. Concepts of Computer Vision

A. Computer Vision Tasks:
- **Image Classification:** Assigning a label to an image.
- **Object Detection:** Locating and classifying objects in an image.
- **Image Segmentation:** Dividing an image into regions.
- **Object Tracking:** Tracking objects in a video.

B. Basics of Images:
- **Pixel:** The smallest unit of a digital image.
- **Resolution:** The number of pixels in an image (width × height).
- **Pixel Value:** The intensity of a pixel.
  - **Grayscale:** 0 (black) to 255 (white).
  - **RGB:** Red (0-255), Green (0-255), Blue (0-255).

C. Types of Images:
- **Grayscale Image:** One channel (intensity).
- **RGB Image:** Three channels (Red, Green, Blue).

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3. Image Features and Convolution Operator

A. Image Features:
- **Edges:** Boundaries between regions.
- **Corners:** Points where edges meet.
- **Textures:** Repeating patterns.

B. Convolution Operator:
- **Definition:** A mathematical operation that applies a filter (kernel) to an image.
- **Purpose:** Extract features like edges, corners, and textures.
- **Process:**
  1. Slide the kernel over the image.
  2. Multiply pixel values with kernel values.
  3. Sum the results to get a new pixel value.
- **Result:** A feature map highlighting specific features.

C. Convolutional Neural Networks (CNN):
- **Definition:** A type of neural network designed for image processing.
- **Layers:**
  - **Convolutional Layer:** Applies convolution to extract features.
  - **Pooling Layer:** Reduces the spatial size (downsampling).
  - **Fully Connected Layer:** Makes the final classification.
- **Kernel:** A small matrix used in convolution.
- **Stride:** The step size of the sliding window.

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4. No-Code AI Tools for Computer Vision

A. Lobe:
- **Website:** www.lobe.ai
- **Purpose:** Build image classification models.
- **Steps:**
  1. Import images.
  2. Label them.
  3. Train the model.
  4. Export and use.

B. Teachable Machine:
- **Website:** teachablemachine.withgoogle.com
- **Purpose:** Train models for images, sounds, and poses.
- **Activity:** Build a Smart Sorter – Classify objects using images.

C. Orange Data Mining:

- **Case Study:** Coral Bleaching Detection
- **Problem:** Detect coral bleaching using images.
- **Steps:**
  1. Load image data.
  2. Preprocess images.
  3. Extract features.
  4. Train a classification model.
  5. Evaluate the model.

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5. Convolutional Neural Networks (CNN)

Architecture:
1. **Input Layer:** The input image.
2. **Convolutional Layer:** Extracts features using kernels.
3. **Pooling Layer:** Reduces dimensions (max pooling).
4. **Flattening Layer:** Converts 2D data to 1D.
5. **Fully Connected Layer:** Classification.
6. **Output Layer:** The final prediction.

Key Concepts:
- **Kernel/Filters:** Extract features.
- **Stride:** Step size of the kernel.
- **Padding:** Adding borders to the image.
- **Activation Function:** ReLU (Rectified Linear Unit).

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. What is computer vision? Give examples.
2. What is a pixel? What is resolution?
3. What is the difference between grayscale and RGB images?
4. What is a convolution operator?

Long Answer (4-5 marks):
1. Explain the architecture of a CNN.
2. Describe the process of convolution in image processing.
3. How can no-code AI tools be used for computer vision projects?

UNIT 6: NATURAL LANGUAGE PROCESSING (NLP)

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1. Introduction to NLP

Definition: A field of AI that enables computers to understand, interpret, and generate human language.

Why NLP is Challenging:
- **Ambiguity:** Words can have multiple meanings.
- **Context:** Meaning depends on context.
- **Sarcasm and Irony:** Hard to detect.
- **Slang and Idioms:** Not literal.

Applications:
- Voice assistants (Siri, Google Assistant).
- Language translation (Google Translate).
- Chatbots and virtual assistants.
- Sentiment analysis.
- Text classification.
- Keyword extraction.
- Auto-generated captions.

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2. Stages of NLP

A. Lexical Analysis:
- Breaking down text into words and phrases (tokens).
- **Tokenisation:** Splitting text into tokens.

B. Syntactic Analysis:
- Analysing grammar and sentence structure.
- **Parsing:** Checking if sentences follow grammar rules.

C. Semantic Analysis:
- Understanding the meaning of words and sentences.
- **Word Sense Disambiguation:** Determining the correct meaning of a word.

D. Logical Analysis:
- Understanding the logical structure of sentences.
- **Inference:** Drawing conclusions.

E. Pragmatic Analysis:
- Understanding the context and purpose.
- **Discourse Analysis:** Understanding the overall meaning.

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3. Chatbots

Definition: A computer program that simulates human conversation.

Types of Chatbots:

A. Script Bots:
- Follow predefined rules and scripts.
- **Example:** Customer service FAQs.
- **Advantages:** Simple, reliable.
- **Disadvantages:** Limited, cannot handle complex queries.

B. Smart Bots:
- Use AI (NLP, ML) to understand and respond.
- **Example:** Siri, Alexa, Google Assistant.
- **Advantages:** Can handle complex queries, learn from interactions.
- **Disadvantages:** More complex to build.

Popular Chatbots to Explore:
- Elizabeth: A simple conversational agent.
- Mitsuku (Kuki): An award-winning chatbot.
- Cleverbot: A chatbot that learns from conversations.

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4. Text Processing

A. Text Normalisation:
- Preparing text for analysis.
- **Steps:**
  - **Tokenisation:** Splitting text into words.
  - **Stopword Removal:** Removing common words (and, the, is).
  - **Stemming:** Reducing words to their root form (e.g., "running" → "run").
  - **Lemmatisation:** Reducing words to their dictionary form (e.g., "better" → "good").

B. Bag of Words (BoW):
- **Definition:** A representation of text that counts the frequency of words.
- **Process:**

  1. Create a vocabulary of unique words.
  2. Count the frequency of each word in the text.
  3. Represent text as a vector of word counts.
- **Example:**
  - Document 1: "The cat sat on the mat."
  - Document 2: "The dog sat on the mat."
  - Vocabulary: {the, cat, sat, on, mat, dog}
  - Document 1 vector: [2, 1, 1, 1, 1, 0]
  - Document 2 vector: [2, 0, 1, 1, 1, 1]

C. TF-IDF (Term Frequency - Inverse Document Frequency):
- **Definition:** A more advanced method that considers word importance.
- **Term Frequency (TF):** How often a word appears in a document.
- **Inverse Document Frequency (IDF):** How rare a word is across documents.
- **Formula:** TF-IDF = TF × IDF
- **Purpose:** Gives higher weight to words that are important in a document.

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5. Sentiment Analysis

Definition: Determining the emotional tone (positive, negative, neutral) of a piece of text.

Applications:
- Social media monitoring.
- Customer feedback analysis.
- Brand reputation monitoring.

Use Case Walkthrough:
- **Problem:** Analyse customer reviews to determine sentiment.
- **Steps:**
  1. Load the dataset (customer reviews with labels).
  2. Preprocess the text (tokenisation, stopword removal).
  3. Convert text to BoW or TF-IDF vectors.
  4. Train a classification model.
  5. Evaluate the model.
  6. Use the model to predict sentiment on new reviews.

Tools:
- Orange Data Mining for visual workflow.
- Python libraries: NLTK, Scikit-learn.

IMPORTANT QUESTIONS

Short Answer (2-3 marks):
1. What is NLP? Why is it challenging?
2. What are the stages of NLP?
3. What is a chatbot? What are the types of chatbots?
4. What is sentiment analysis?

Long Answer (4-5 marks):
1. Explain the Bag of Words and TF-IDF models.
2. Describe the steps involved in text processing.
3. How can NLP be used in real-life applications?

UNIT 7: ADVANCED PYTHON (PRACTICAL)

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1. Jupyter Notebook

What is Jupyter Notebook?
- An interactive web-based environment for writing and running code.
- Supports multiple languages (Python, R, Julia).

Key Features:
- **Cells:** Code cells, Markdown cells, Raw cells.
- **In-line Output:** Results are displayed below the code.
- **Visualisation:** Supports plots and charts.

Installation:
- Using Anaconda Navigator.
- Using pip: `pip install jupyter`

Running Jupyter Notebook:
- Command: `jupyter notebook`

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2. Python Basics

A. Variables:
- Containers for storing data.
- No need to declare the type.

- **Example:** `x = 5`, `name = "John"`

B. Data Types:
- **Integer:** Whole numbers (e.g., 5, -3, 0).
- **Float:** Decimal numbers (e.g., 3.14, -2.5).
- **String:** Text (e.g., "Hello", 'World').
- **Boolean:** True or False.
- **List:** Ordered collection (e.g., [1, 2, 3]).
- **Tuple:** Immutable ordered collection (e.g., (1, 2, 3)).
- **Dictionary:** Key-value pairs (e.g., {"name": "John", "age": 25}).

C. Operators:
- **Arithmetic:** +, -, *, /, %, ** (exponent), // (floor division).
- **Comparison:** ==, !=, >, <, >=, <=.
- **Logical:** and, or, not.
- **Assignment:** =, +=, -=, *=, /=.

D. Control Structures:
- **if-else:**
  ```python
  if condition:
      statement
  elif condition:
      statement
  else:
      statement
  ```
- **for loop:**
  ```python
  for i in range(5):
      print(i)
  ```
- **while loop:**
  ```python
  i = 0
  while i < 5:
      print(i)
      i += 1
  ```

E. Functions:
- **Definition:**
  ```python
  def function_name(parameters):
  ```

      # statements
      return value
  ```
- **Example:**
  ```python
  def add(a, b):
      return a + b
  ```

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3. Python Libraries

A. Numpy:
- Numerical computing.
- **Key Features:** Arrays, mathematical functions.
- **Installation:** `pip install numpy`
- **Example:**
  ```python
  import numpy as np
  arr = np.array([1, 2, 3, 4, 5])
  mean = np.mean(arr)
  ```

B. Pandas:
- Data manipulation and analysis.
- **Key Features:** DataFrames, reading CSV files.
- **Installation:** `pip install pandas`
- **Example:**
  ```python
  import pandas as pd
  df = pd.read_csv('filename.csv')
  print(df.head(10))
  ```

C. Matplotlib:
- Data visualisation.
- **Key Features:** Line charts, scatter plots, bar charts.
- **Installation:** `pip install matplotlib`
- **Example:**
  ```python
  import matplotlib.pyplot as plt
  x = [2, 9, 8, 5, 6]
  y = [5, 10, 3, 7, 18]

  plt.scatter(x, y)
  plt.show()
  ```

D. Scikit-learn:
- Machine learning.
- **Key Features:** Model selection, preprocessing, evaluation.
- **Installation:** `pip install scikit-learn`
- **Example:**
  ```python
  from sklearn.model_selection import train_test_split
  ```

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4. Suggested Practical Programs

1. **Add Elements of Two Lists:**
   ```python
   list1 = [1, 2, 3]
   list2 = [4, 5, 6]
   result = [a + b for a, b in zip(list1, list2)]
   print(result)  # [5, 7, 9]
   ```

2. **Calculate Mean, Median, Mode using Numpy:**
   ```python
   import numpy as np
   from scipy import stats
   data = [1, 2, 2, 3, 4, 5, 6]
   mean = np.mean(data)
   median = np.median(data)
   mode = stats.mode(data)
   print(f"Mean: {mean}, Median: {median}, Mode: {mode}")
   ```

3. **Display Line Chart from (2,5) to (9,10):**
   ```python
   import matplotlib.pyplot as plt
   x = [2, 9]
   y = [5, 10]
   plt.plot(x, y)
   plt.show()
   ```
   ```

4. **Display Scatter Chart:**
   ```python
   import matplotlib.pyplot as plt
   points = [(2,5), (9,10), (8,3), (5,7), (6,18)]
   x = [p[0] for p in points]
   y = [p[1] for p in points]
   plt.scatter(x, y)
   plt.show()
   ```

5. **Read CSV File and Display 10 Rows:**
   ```python
   import pandas as pd
   df = pd.read_csv('filename.csv')
   print(df.head(10))
   ```

6. **Read CSV File and Display Information:**
   ```python
   import pandas as pd
   df = pd.read_csv('filename.csv')
   print(df.info())
   ```

7. **Read and Display an Image:**
   ```python
   import cv2
   import matplotlib.pyplot as plt
   img = cv2.imread('image.jpg')
   img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
   plt.imshow(img_rgb)
   plt.show()
   ```

8. **Read an Image and Identify its Shape:**
   ```python
   import cv2
   img = cv2.imread('image.jpg')
   shape = img.shape
   print(f"Image shape: {shape}")  # (height, width, channels)
   ```

KEY TERMS AND DEFINITIONS

AI (Artificial Intelligence): The simulation of human intelligence processes by machines.

ML (Machine Learning): A subset of AI where machines learn from data.

DL (Deep Learning): A subset of ML using neural networks with many layers.

Supervised Learning: Learning from labelled data.

Unsupervised Learning: Learning from unlabelled data.

Reinforcement Learning: Learning through trial and error.

Neural Network: A computing system inspired by the human brain.

CNN (Convolutional Neural Network): A neural network designed for image processing.

Computer Vision: Enabling machines to interpret visual information.

NLP (Natural Language Processing): Enabling machines to understand human language.

Sentiment Analysis: Determining the emotional tone of text.

Chatbot: A program that simulates human conversation.

Bag of Words: A text representation model that counts word frequencies.

TF-IDF: A text representation model that considers word importance.

Confusion Matrix: A table summarising classification model performance.

Precision: Correct positive predictions / total positive predictions.

Recall: Correct positive predictions / actual positives.

F1 Score: Harmonic mean of precision and recall.

Accuracy: Correct predictions / total predictions.

Train-Test Split: Dividing data into training and testing sets.

Ethical Framework: Guidelines for responsible AI development.

Bias: Systematic error in AI models.

No-Code AI: Building AI models without programming.

Orange Data Mining: A visual programming tool for data analysis.

Pixel: The smallest unit of a digital image.

Kernel: A filter used in convolution.

Jupyter Notebook: An interactive coding environment.

Lobe: A no-code tool for building image classification models.

Teachable Machine: A no-code tool for training models.
